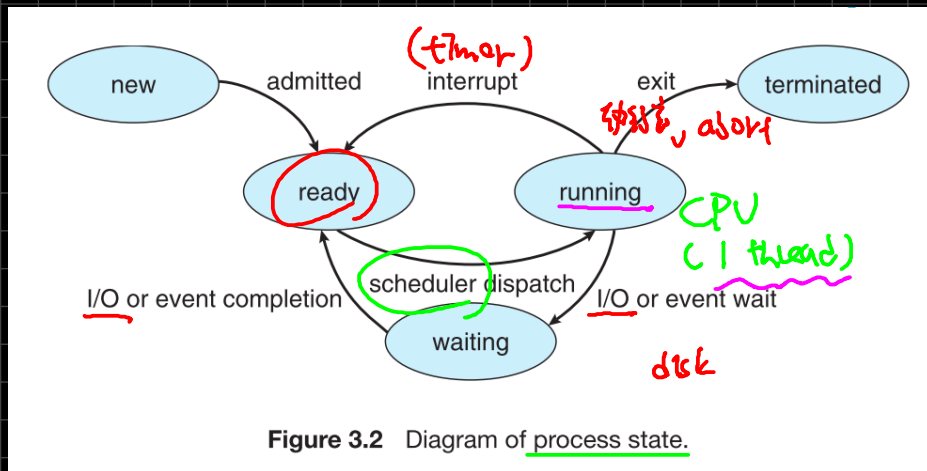


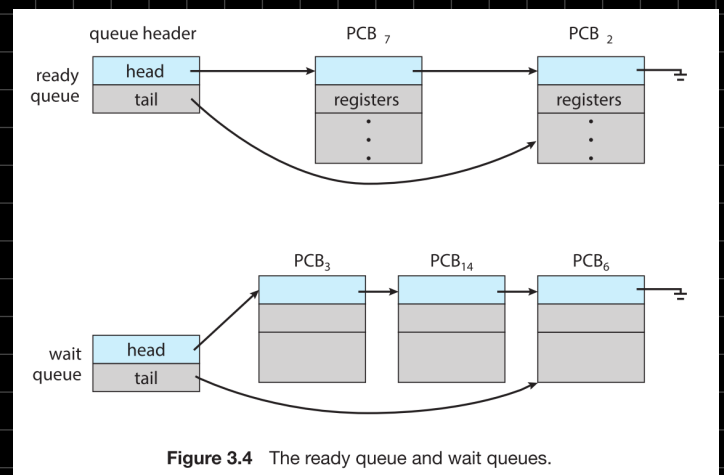
process state

都在 process 層級

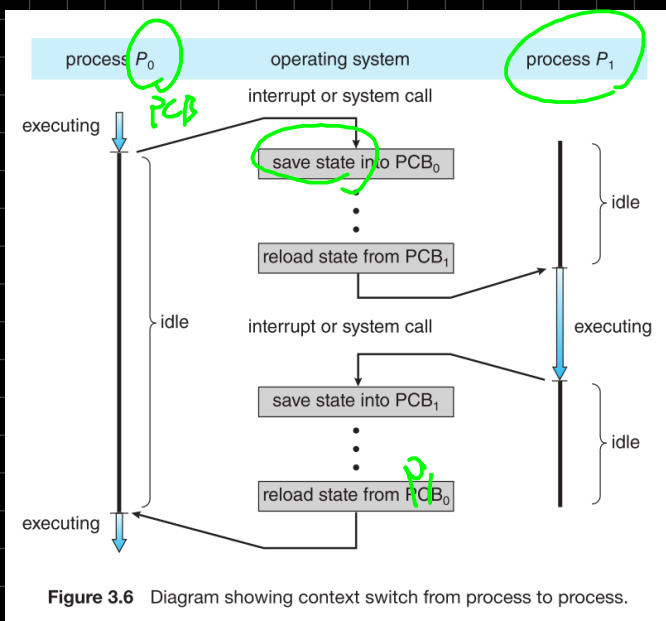


process → new OS → ready queue
create

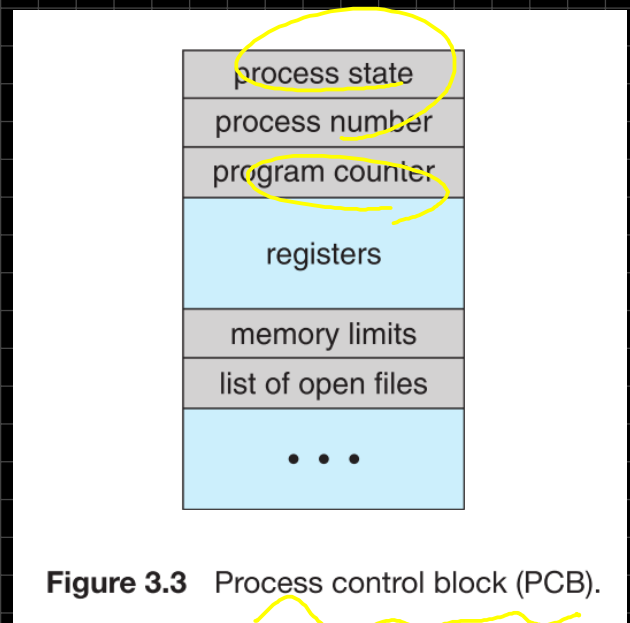
1 process = 1 thread



context switch



(1) store old state
(2) load new state



process priority := 每個 process 會被賦予一個值，代表緊迫程度。

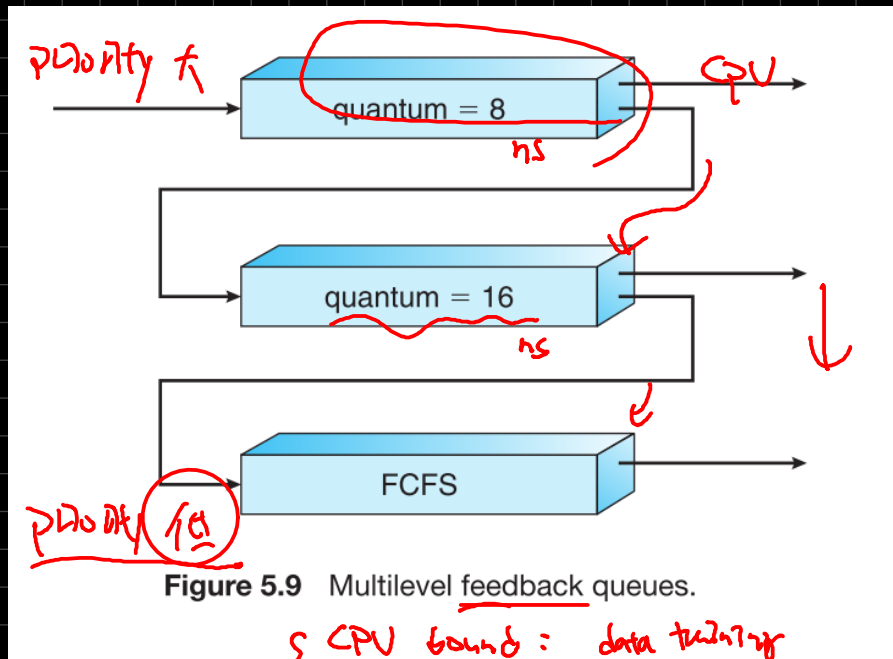
CPU scheduler 會優先分配給 priority 最高的 process (數字越小，priority 越大)

$p_1 = 2$ (大)
 $p_2 = 100$ (小)

time quantum := 1 個 process 在 CPU 上最大執行的時間

ch4
CPU
scheduling

Multilevel Feedback Queue, MLFQ



SJF

RR

perhaps SJF

CPU bound: data transferring
I/O bound: ex. mouse

- The method used to determine when to upgrade a process to a higher-priority queue
- The method used to determine when to demote a process to a lower-priority queue

前 queue 中無法在規定的 time slice 內完成，它就會被移至更低優先權的佇列中

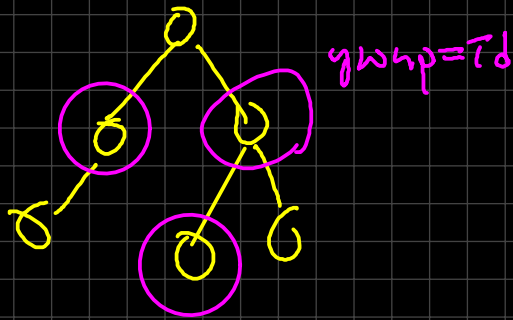
13. Which of the following statements are correct?

- (a) A process with multiple user threads and one kernel thread (many-to-one) cannot run in parallel on multiprocessors.
- (b) Threads belonging to a process share one program counter.
- (c) Operating system schedule kernel threads and user threads as well.
- (d) Most operating systems, such as Solaris and Windows, boost the thread priority when it runs out of time quantum.
- (e) Most operating systems, such as Solaris and Windows, boost the thread priority when it returns from an I/O request.

I/O bound process

CPU ↓

process group Td



thread model

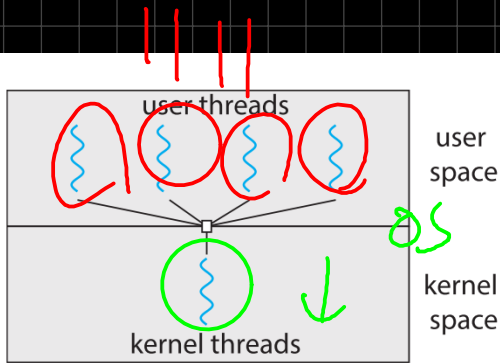


Figure 4.7 Many-to-one model.

concurrent

parallel (multi-core)

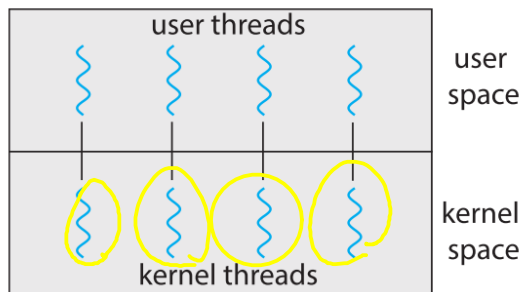
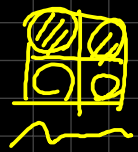


Figure 4.8 One-to-one model.

1:1



context overhead
x efficiency

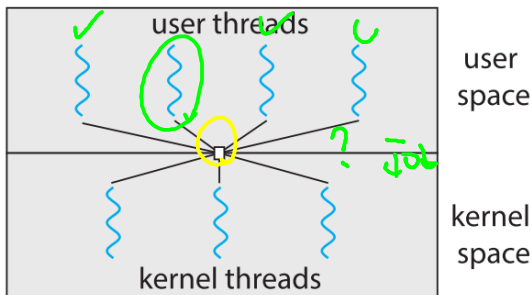


Figure 4.9 Many-to-many model.

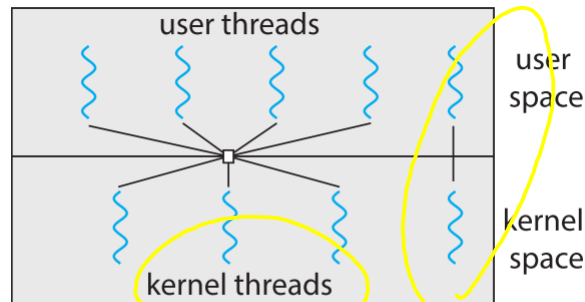
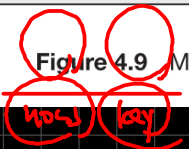


Figure 4.10 Two-level model.

* 沒有 one to many model, 因為 thread 為基本單位, 一個 user thread 不能再切割, 必須到底到一個 kernel thread